Supporting Information

Decomposition of Proteins into Dynamic Units from Atomic Cross-Correlation Functions

Paolo Calligari,* Marco Gerolin, Daniel Abergel,* and Antonino Polimeno*

E-mail: paolo.calligari@unipd.it; daniel.abergel@ens.fr; antonino.polimeno@unipd.it

Downloaded from pubs.acs.org/jctcce/article-supplement/691992/pdf/ct6b00702_si_001/ by guest on 24 October 2026

1 Settings for the Affinity Propagation algorithm

Prior to its application, the Affinity Propagation (AP) algorithm was parameterized with respect to the dissimilarity matrix at hand in order to obtain the highest silhouette score $\overline{\mathcal{S}}$ from the clustering procedure. As suggested by Frey and Dueck, and other authors,[1,2] a single common preference value P is used for all the nodes. However, since the value of P impacts the result of the clustering algorithm,[2] it was necessary to determine its best choice for initialisation of the AP algorithm. This was achieved by repeating the clustering procedure for initial values of P, that were common to all the nodes and sampled the interval between the minimal to the maximal distances in the similarity matrix by steps of 50 ns^2 . P was assigned a negative value to follow the prescriptions given in the original work by Frey *et al.*[2] During this procedure, the quality of the clustering was assessed by the silouhette score $\overline{\mathcal{S}}$ (see manuscript). The influence of P on the number of clusters and on the silhouette score $\overline{\mathcal{S}}$ obtained from the HP35 **S** matrix is illustrated in Figure S1. The highest quality of the clustering protocol, i.e. the maximal value of $\overline{\mathcal{S}}$, is achieved for $P \leq -2 \cdot 10^8$. Below this threshold, the number of clusters remains constant and equal to two . Alternatively, for $P > -200$ ns^2, $\overline{\mathcal{S}}$ decreases of about 30%, whilst the number of clusters rapidly increases. It

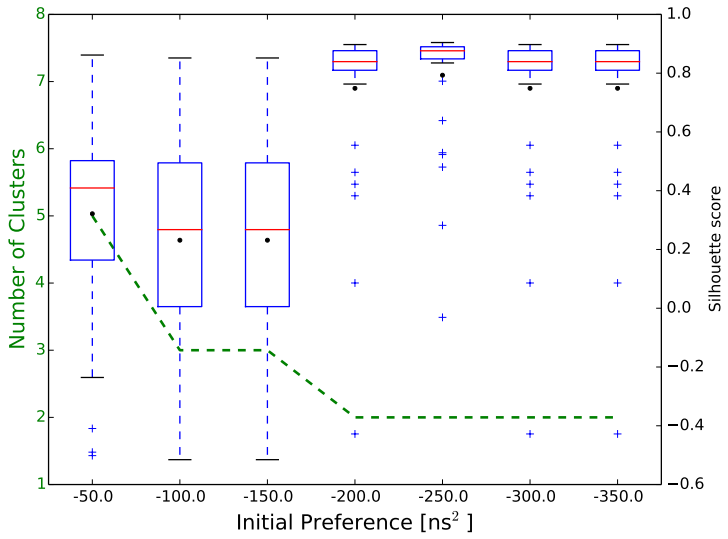


Figure S1: HP35. Number of clusters detected by the AP algorithm and corresponding silhouette overall score $\overline{\mathcal{S}}$ as a function of the initial preference parameter. Error bars indicate the standard deviation of all $\mathcal{S}(i)$ found in HP35 for each initial preference value.Maybe plot the number of clusters as discontinuous line segments, since they are integers

is worth noting that our protocol yields well-defined clusters, as suggested by the relatively small dispersion of the set $\{\mathcal{S}(i)\}$ (indicated by the "error bars" in box-plot shown in Figure S1). Therefore, based on this analysis, we decided to set $P = -250$ ns^2, which gives the largest value of $\overline{\mathcal{S}}$ and the smallest dispersion in the set $\{\mathcal{S}(i)\}$. This choice was used in all subsequent analyses in this work.

Downloaded from pubs.acs.org/jctcce/article-supplement/691992/pdf/ct6b00702_si_001/ by guest on 24 October 2026

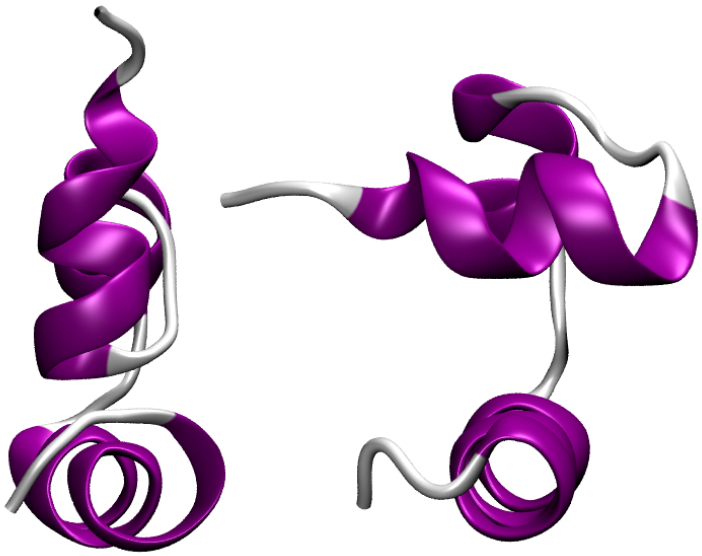


Figure S2: Snapshots from the MD simulations of HP35 show the significant mobility of the N-terminal α-helix.

Downloaded from pubs.acs.org/jctcce/article-supplement/691992/pdf/ct6b00702_si_001/ by guest on 24 October 2026

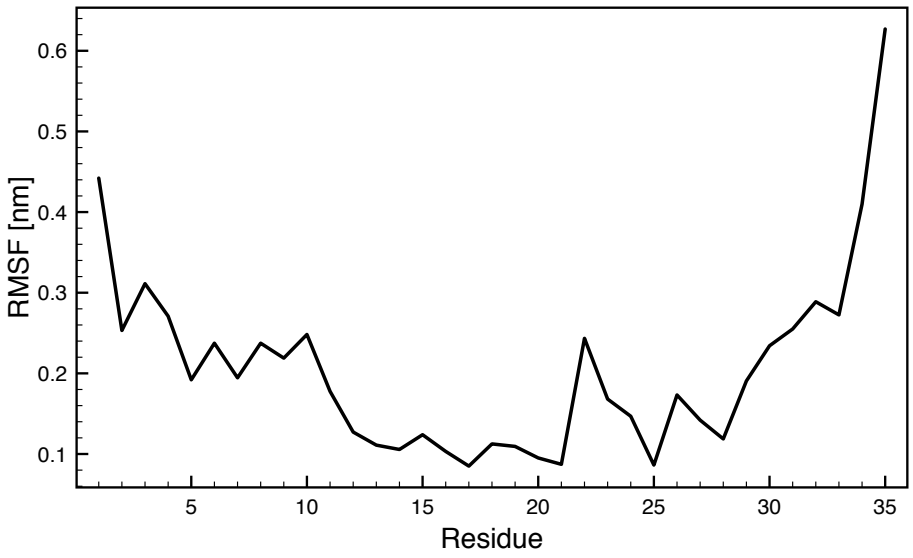


Figure S3: Root mean square fluctuations in HP35. Average atom rmsf are plotted for each residue.

References

(1) Vlasblom, J.; Wodak, S. J. *BMC Bioinformatics* **2009**, *10*, 99.

(2) Frey, B.; Dueck, D. *Science* **2007**, *315*, 972.

Downloaded from pubs.acs.org/jctcce/article-supplement/691992/pdf/ct6b00702_si_001/ by guest on 24 October 2026